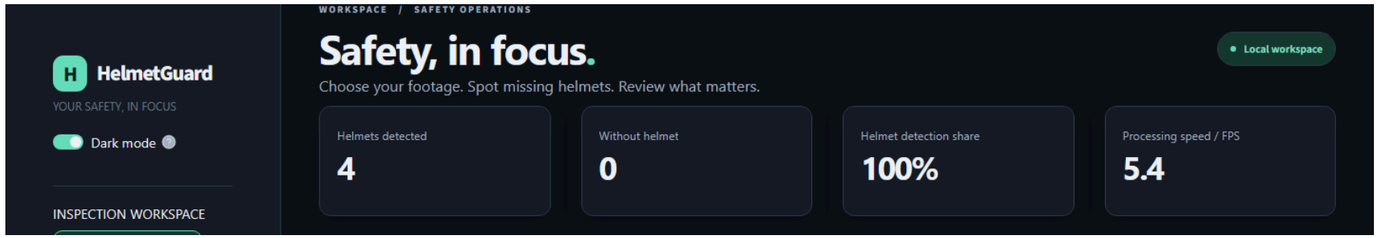
HelmetGuard





AI-powered helmet monitoring for workplace safety

Developed a computer vision prototype that detects helmets and bare heads in images, uploaded videos, and live camera feeds. The dashboard highlights detections and flags missing helmets for review.

Model Architecture

- Ultralytics YOLO object detection
- Explicit Hardhat and NO-Hardhat classes
- Frame processing with OpenCV
- Adjustable confidence threshold
- Bounding boxes and review alerts
- Per-frame counts and processing FPS
- Streamlit dashboard with light/dark themes

Technologies Used

Python, Ultralytics YOLO, OpenCV, Streamlit

Prototype • Camera performance requires field validation